Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)
Department of Computer Science and Engineering

CAT Assignment 1

Degree & Branch:	M.Tech (Integrated) Computer Science and Engineering
Semester:	V
Subject Code & Name:	ICS1502 - Introduction to Machine Learning
Academic Year:	2025–2026 (Odd Semester)
Batch:	2023–2028
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1. Programming Assignment

1.1 Regression: Mobile Phone Price Prediction

Aim: To implement and analyze a Linear Regression model for predicting mobile phone prices using both closed-form and gradient descent methods, with and without L2 regularization, and to evaluate their performance under various preprocessing settings.

Objectives

- To construct data and label matrices for linear regression.
- To implement regression using closed-form and gradient descent solutions.
- To study the effect of regularization (L2 penalty) on model generalization.
- To compare model performance before and after data standardization.
- To interpret model weights and understand feature importance.

Mathematical Description

Given a dataset (X, y) with m samples and n features, the linear regression model predicts:

$$\hat{y} = X\theta$$

The closed-form solution (Ordinary Least Squares) is:

$$\theta = (X^T X)^{-1} X^T y$$

For Ridge Regression (L2 regularization):

$$\theta = (X^T X + \lambda I)^{-1} X^T y$$

For Gradient Descent:

$$\theta := \theta - \alpha \frac{1}{m} X^T (X\theta - y)$$

where α is the learning rate and λ controls the regularization strength.

Results

Exploratory Data Analysis (EDA):

- Checked for missing values and feature distributions.
- Generated correlation heatmap to identify feature relationships.

Pairplot and Correlation Heatmap:

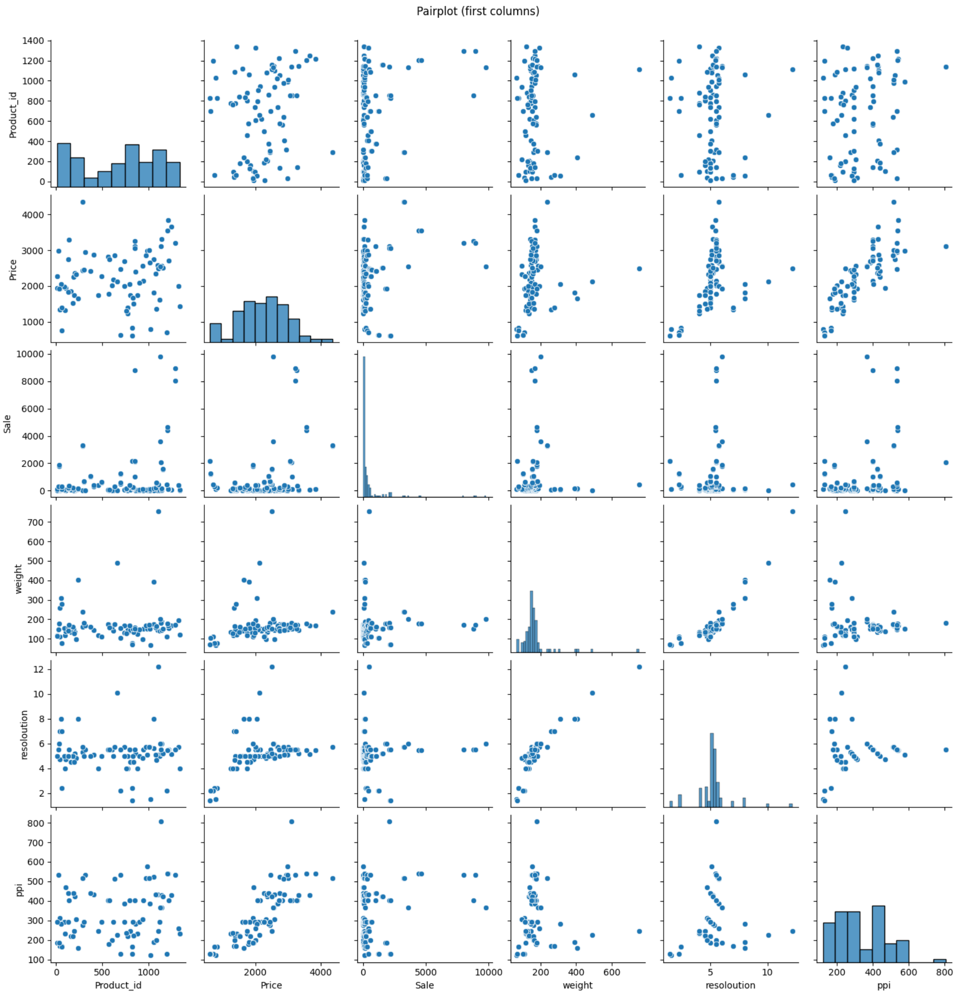

Figure 1: Pairplot of features in Mobile Price dataset

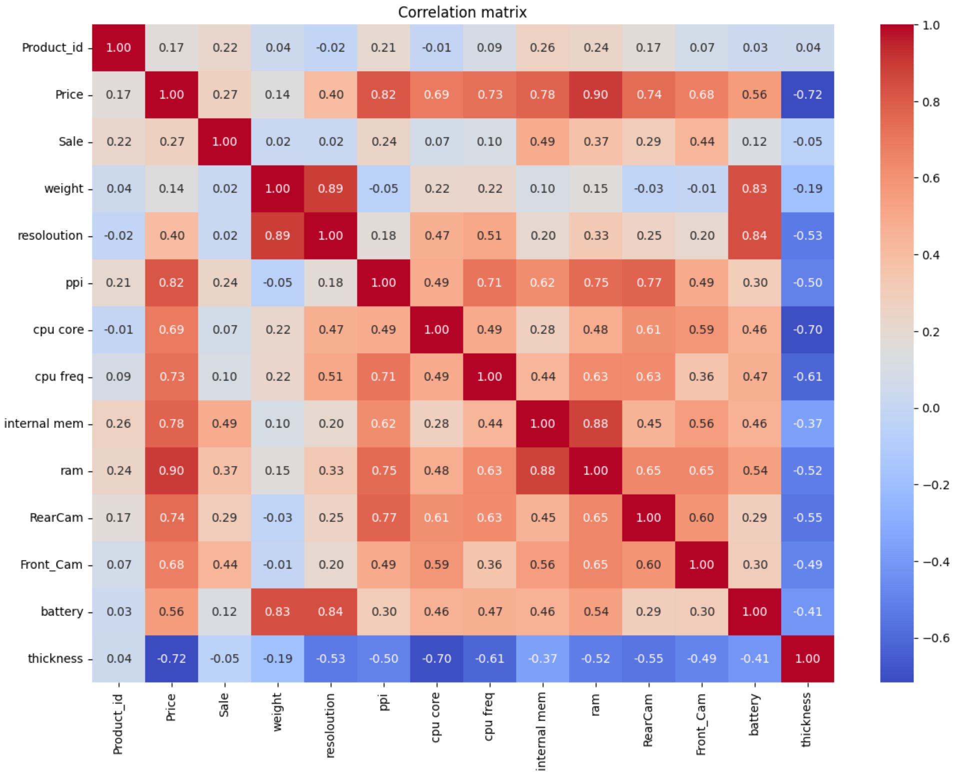

Figure 2: Feature correlation heatmap

Performance Metrics:

- **Closed-form Solution:**
  MSE = 23062.23, RMSE = 151.86, MAE = 130.32, R^2 = 0.9593
- **Gradient Descent:**
  MSE = 8.2462e+10, RMSE = 287162.01, MAE = 264011.00, R^2 = -145463.13
- **Ridge (Closed-form, λ=10):**
  Test RMSE = 161.09
- **Ridge (Gradient Descent, λ=10):**
  Test RMSE = 1909.43

Comparison Across λ Values:

Lambda	No-Scale RMSE	Scale RMSE	No-Scale R^2	Scale R^2
0.00	151.86	151.86	0.9593	0.9593
0.01	151.86	151.87	0.9593	0.9593
0.10	151.87	151.92	0.9593	0.9593
1.00	152.01	152.49	0.9592	0.9589
10.00	154.24	161.09	0.9580	0.9542
100.00	166.65	230.26	0.9510	0.9065

Predicted vs Actual Plots:

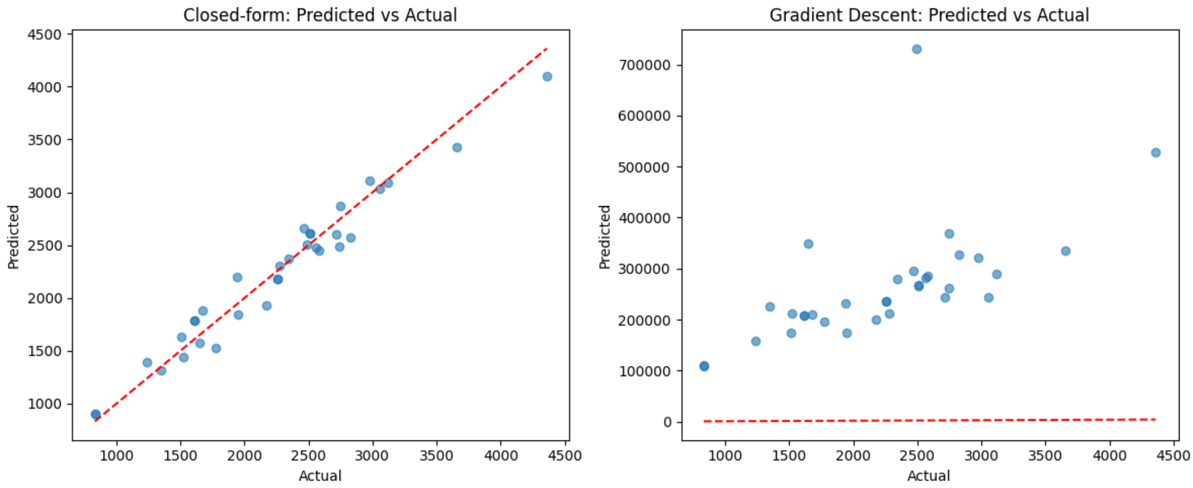

Figure 3: Predicted vs Actual Plot

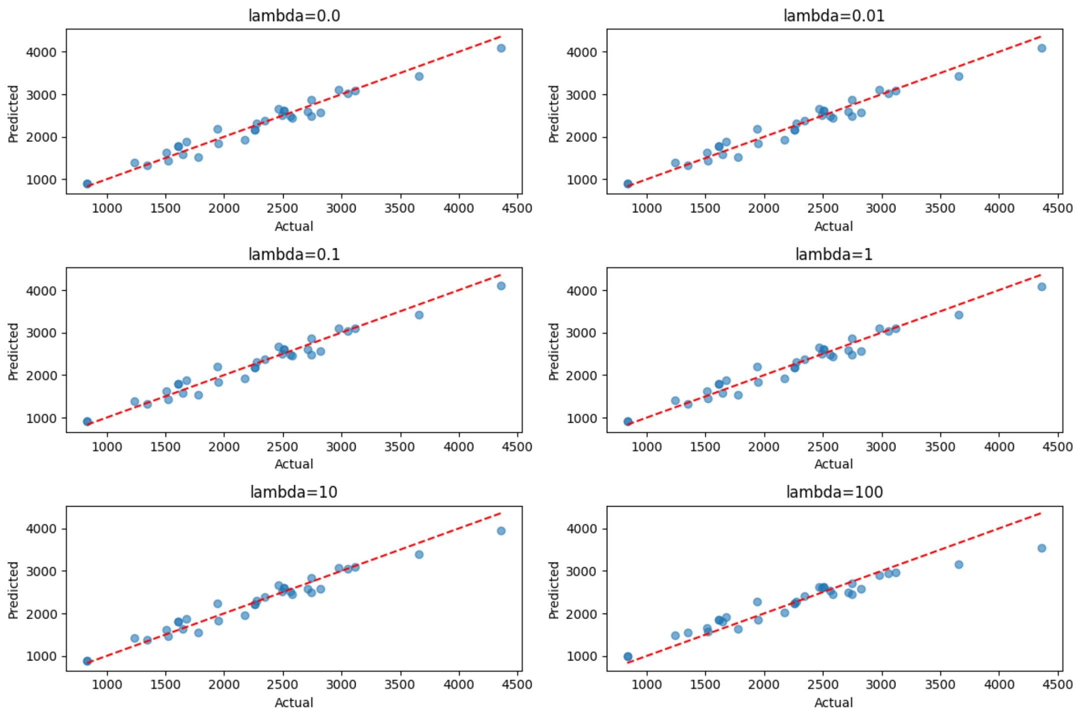

Figure 4: Predicted vs Actual for Ridge Regression with different λ values

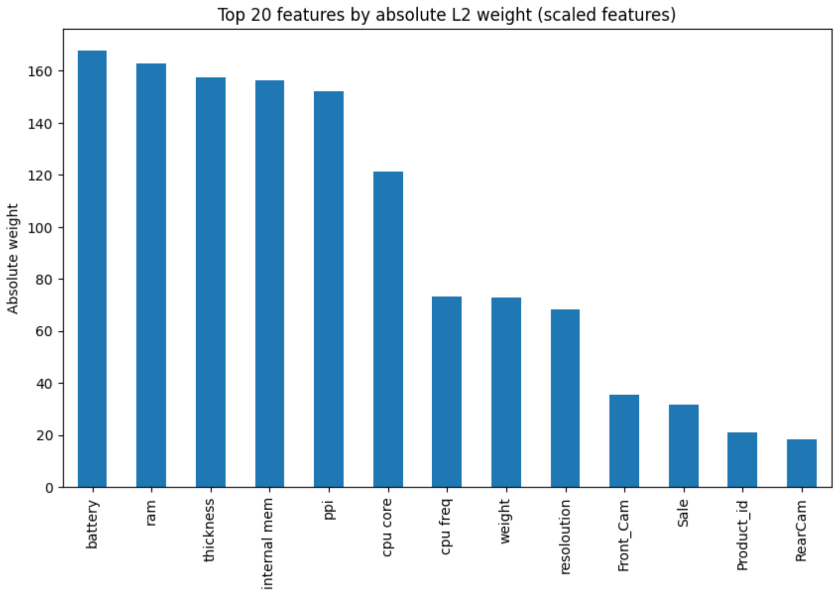

Figure 5: Feature Importance Plot

Inference

- The closed-form linear regression performed well with a high R^2 (0.9593), showing strong predictive power.
- Gradient Descent failed to converge due to improper learning rate or lack of normalization.
- Ridge regularization slightly reduced RMSE, preventing overfitting for larger λ values.
- Excessively large λ degraded model performance, emphasizing the need for regularization tuning.
- Feature importance analysis showed that certain features had strong linear relationships with price, consistent with correlation heatmap findings.
- Standardization of data improved the stability of results for smaller λ values.

1.2 Linear Classification: Bank Note Authentication

Aim: To fit a linear classification model for banknote authentication using logistic regression and neural networks, evaluate L2 regularization, and analyze model robustness under outlier injection.

Objectives

- To train and test a linear classifier on the Bank Note Authentication dataset.
- To compare model accuracy with and without L2 regularization.
- To analyze the effect of outliers on model performance.
- To visualize 3D decision boundaries using key features.

Mathematical Description

The logistic regression model predicts the probability:

$$P(y = 1|x) = \sigma(w^T x + b)$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$.

The cost function with L2 regularization:

$$J(w) = -\frac{1}{m}\sum_{i=1}^{m}[y^{(i)}\log(\hat{y}^{(i)}) + (1 - y^{(i)})\log(1 - \hat{y}^{(i)})] + \frac{\lambda}{2m}\|w\|^2$$

Results

Exploratory Data Analysis:

- Dataset was clean and well-balanced across classes.
- Features displayed strong linear separability, as seen in pairplots.

Pairplot and Heatmap:

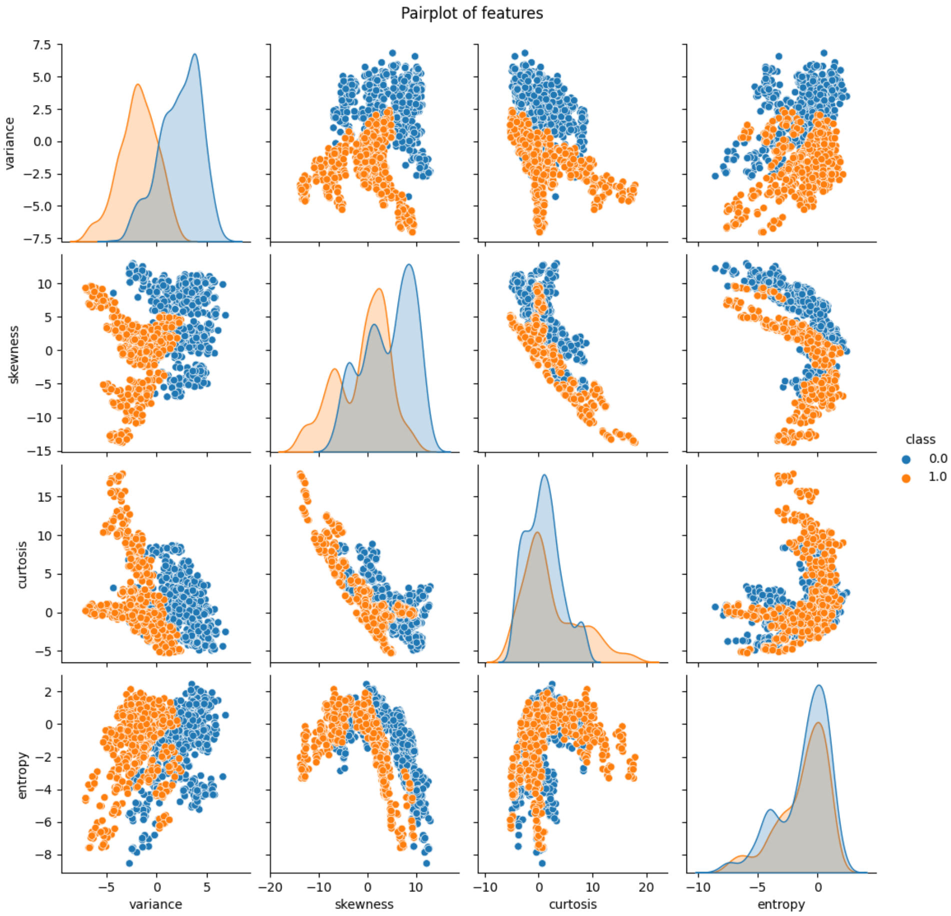

Figure 6: Feature pairplot for Bank Note Authentication dataset

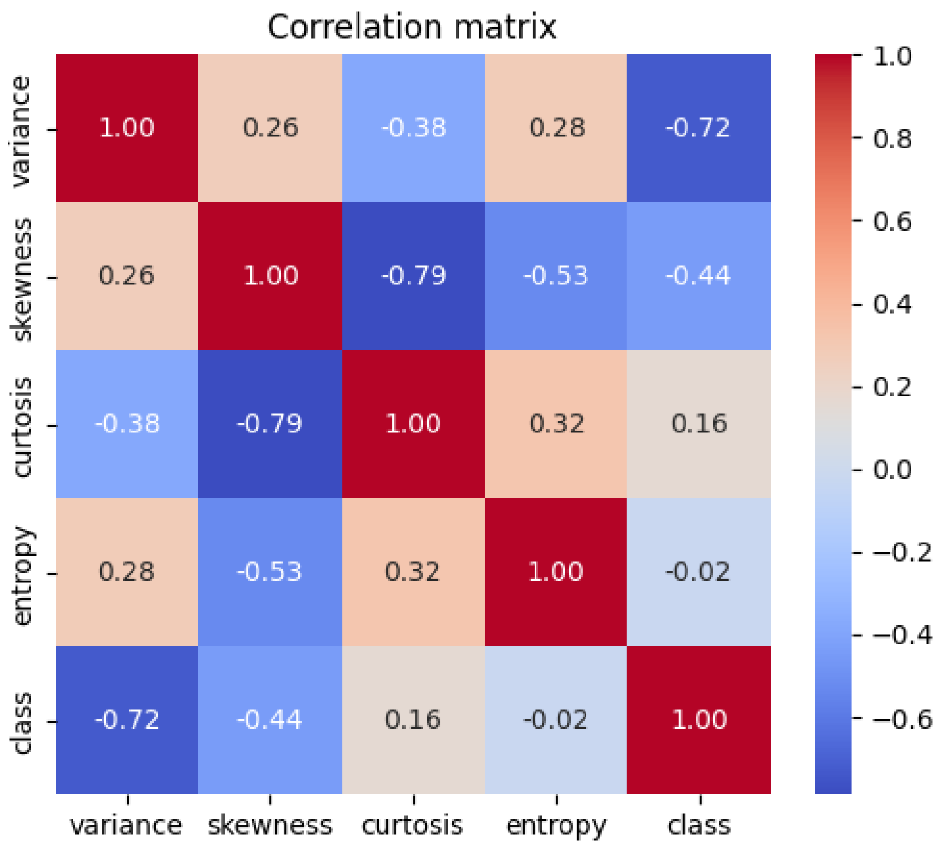

Figure 7: Correlation heatmap of features

Model Performance:

Table 2: Performance Comparison of Models

Model	Train Acc	Test Acc	Precision / Recall / F1
No Regularization	0.9909	0.9854	0.98–0.99
L2 (C=1.0)	0.9845	0.9709	0.97
MLP (1 layer)	1.0000	1.0000	1.00

Feature Importance Ranking:

Feature	Importance Score
Skewness	4.8089
Variance	4.6579
Curtosis	4.3977
Entropy	0.2552

Effect of Outliers:

Model	Train Accuracy	Test Accuracy
Clean Model	0.9845	0.9709
Outlier Model	0.8441	0.8182

Table 4: Impact of outlier injection on model performance

Confusion Matrices:

Table 5: Comparison of Classification Metrics between Clean and Outlier Models

Metric	Class	Clean Model	Outlier Model	Difference (Δ)
Precision	0.0	1.00	0.86	-0.14
	1.0	0.94	0.78	-0.16
Recall	0.0	0.95	0.81	-0.14
	1.0	1.00	0.83	-0.17
F1-Score	0.0	0.97	0.83	-0.14
	1.0	0.97	0.80	-0.17
Accuracy	–	0.97	0.82	-0.15
Macro Avg	–	0.97	0.82	-0.15
Weighted Avg	–	0.97	0.82	-0.15

3D Visualization of Decision Boundaries:

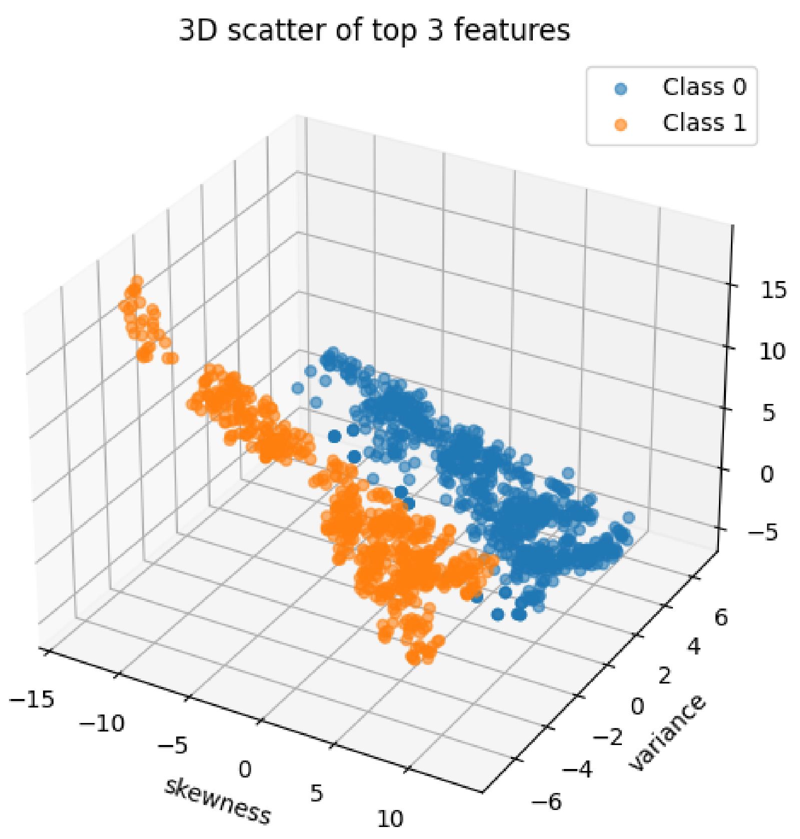

Figure 8: 3D visualization using top three important features

Inference

- The dataset was linearly separable; hence, linear models achieved high accuracy.
- Without regularization, the classifier achieved 98.5% test accuracy, showing minimal overfitting.
- With L2 regularization, accuracy dropped slightly but model stability improved.
- The MLP model achieved perfect accuracy, confirming non-linear models' superior fitting capacity.
- Feature ranking indicated *skewness*, *variance*, and *curtosis* as the most discriminative attributes.
- Injecting outliers drastically reduced accuracy (from 0.97 to 0.82), revealing that linear models are sensitive to anomalous data.
- The decision boundary visualizations confirmed that most samples were well-separated in 3D space.

Learning Outcomes

- Understood matrix-based implementation of linear regression.
- Gained practical insight into gradient descent convergence behavior.
- Learned to tune and interpret regularization effects on bias–variance trade-off.
- Acquired experience in evaluating linear classifiers using performance metrics and confusion matrices.
- Observed the impact of outliers and importance of data quality in supervised learning.
- Enhanced understanding of data visualization and feature interpretation through pairplots, heatmaps, and 3D projections.

GitHub Repository

The complete source code for this assignment is available in the following repository: https://github.com/R-Jayasree/Machine-Learning